AI in the Family

A practical guide for parents: what generative AI is, how it works, and how to use it responsibly at home and at school

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github.com/iportilla/about-me

https://github.com/iportilla/BSD

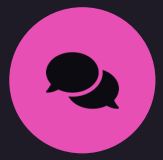
Workshop goals

By the end of this workshop, you will be able to:



Understand

what generative AI is and how it works, without jargon



Ask better

write clear instructions to get better answers



Guide

how your children use AI at school and at home



Take home

a Practical Family Guide for using ChatGPT

Decades of experience in AI and education





University of Colorado Boulder

COLORADO SCHOOL OF MINES | Computer Science

INNOVATION CENTER of ST. VRAIN VALLEY SCHOOLS

github.com/iportilla/about-me

Agenda

1. Welcome: how do we feel about AI?
2. What is generative AI and how does it work? + live demo
3. How to ask good questions
4. **Our kids and AI: school, risks, and opportunities**
5. Hands-on exercise in pairs
6. Can we trust AI? Hallucinations, privacy, and scams
7. AI in everyday life and the Practical Family Guide

A recent example with Artificial Intelligence (AI)

Who wrote this message?

Dear Teacher,

My son has a doctor's appointment and will not be able to attend school next Friday. Could you please let us know which assignments he needs to complete for next week?

Thank you very much.

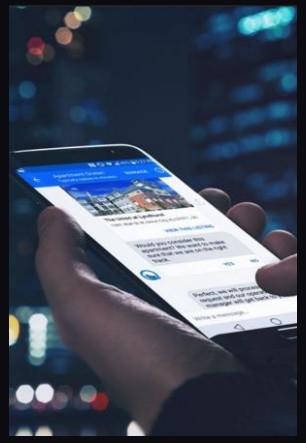
Ask the audience: did a person or AI write this? Does it matter?

How do we feel about AI?



Fearful Distrustful Fascinated Informed

Exposed Anxious Curious Excited Well served

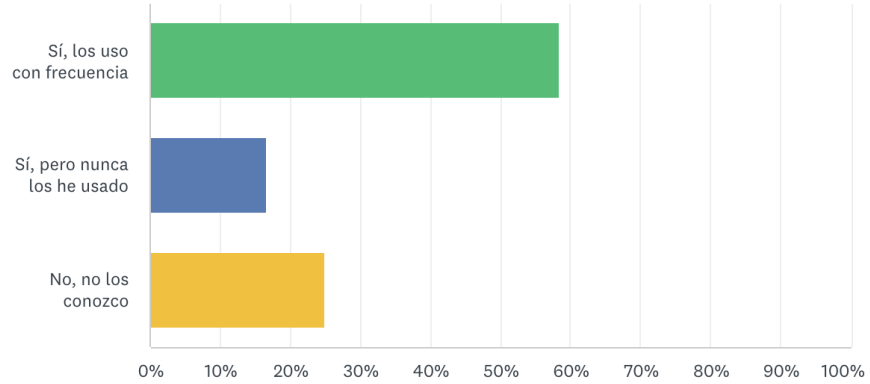


All of these feelings are valid. Today's goal is to move one step to the right: from worry to informed confidence.

Survey: Do you know ChatGPT?

¿Has escuchado hablar de ChatGPT u otros chatbots con inteligencia artificial?

Answered: 12 Skipped: 0

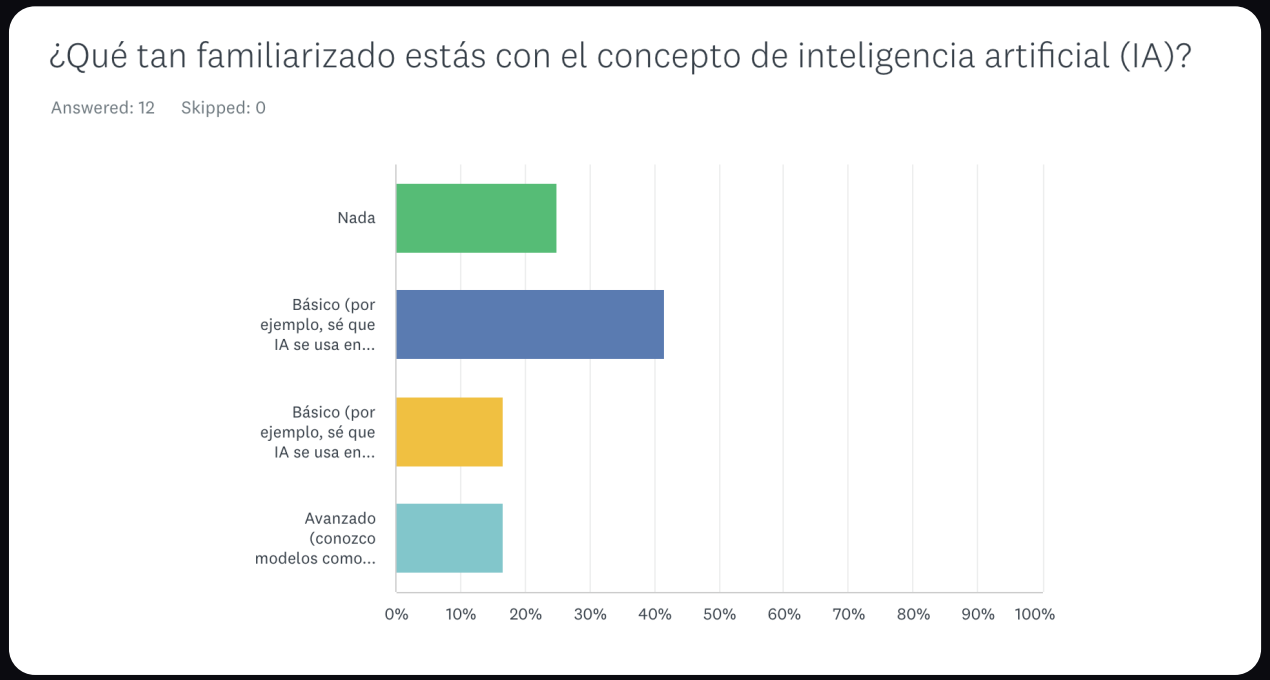


6 in 10

already use chatbots often, but 1 in 4 has never heard of them.

Pre-workshop survey (in Spanish), 12 responses

Survey: How much do you know about AI?

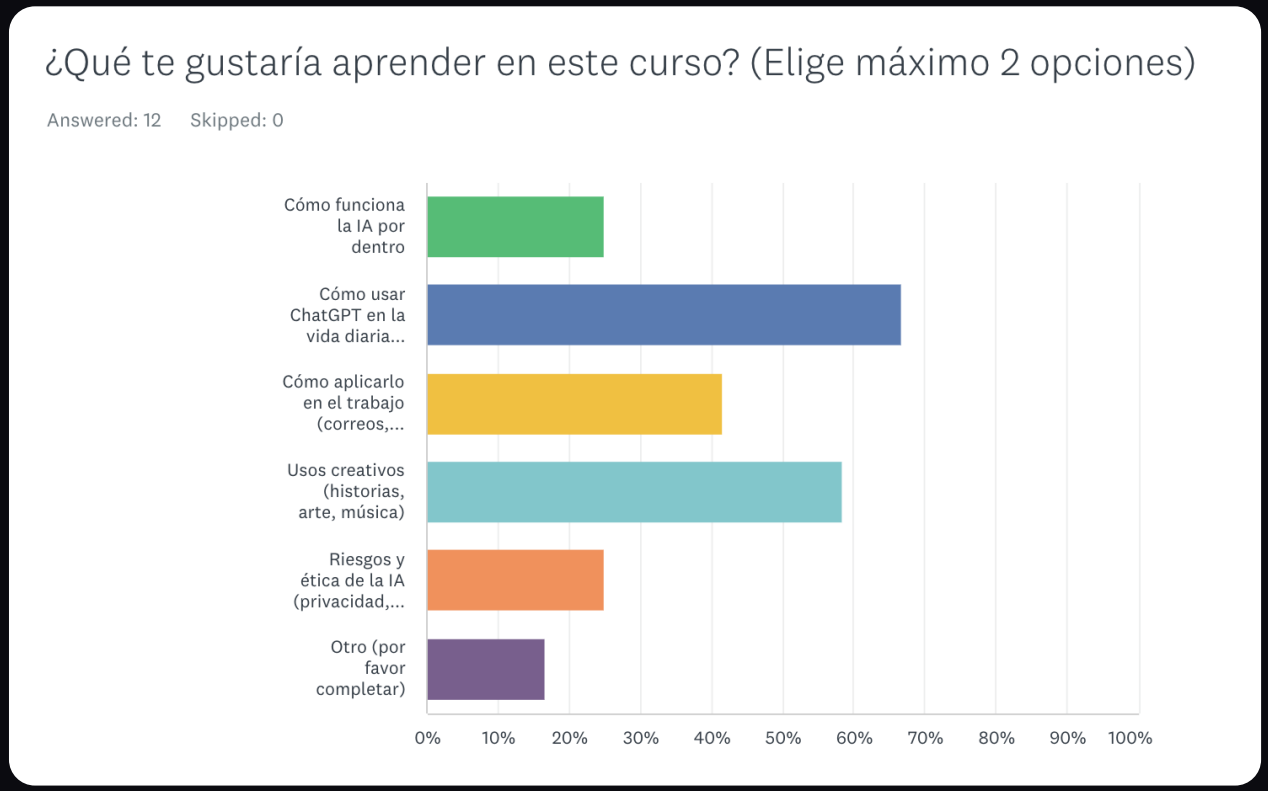


1 in 4

say they know nothing about AI. We start from zero, without jargon.

Pre-workshop survey (in Spanish), 12 responses

Survey: What would you like to learn?

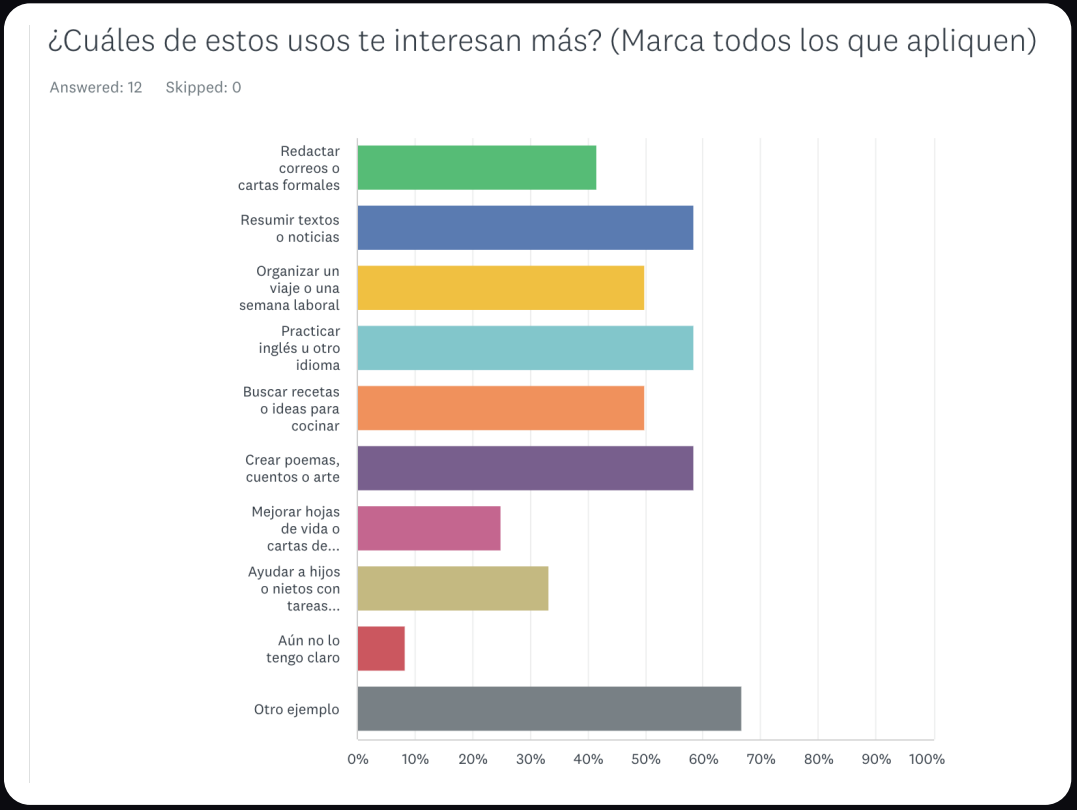


2 in 3

want to learn to use ChatGPT in daily life; creative uses follow close behind.

Pre-workshop survey (in Spanish), 12 responses

Survey: Which uses interest you most?



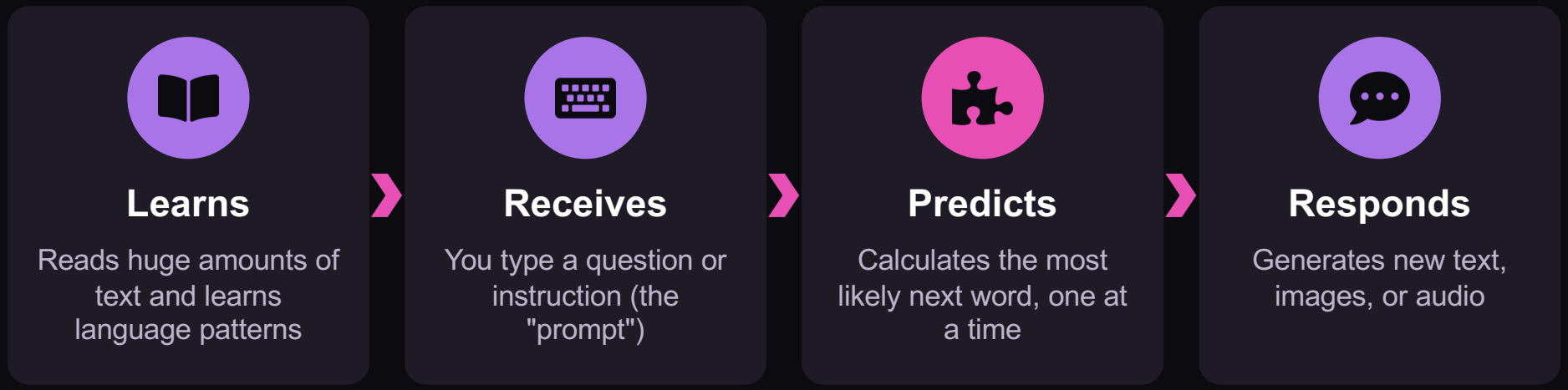
More than half

want to summarize texts, practice languages, and create stories or art.

Pre-workshop survey (in Spanish), 12 responses

How does generative AI work?

It's like your phone's autocomplete, but trained on millions of books and web pages.



Key idea: AI predicts words; it doesn't know the truth. That's why it sounds confident even when it's wrong.

Live demo



- Open ChatGPT on a phone, in English or Spanish
- Talk to it using voice mode
- Ask it to write the email to the teacher
- Ask it to explain "as if I were 10 years old"

No computer needed: a phone is enough.

02

How to ask good questions

Prompt engineering: writing clear instructions that guide AI toward the answer we need.

The 4 parts of a good prompt

Goal

What do you want the AI to do?

"Explain how to solve equivalent fractions…"

Context

Why do you need it and who is involved?

"…so I can help my 10-year-old daughter with her homework."

Expectations

How do you want the answer?

"Use simple words and a pizza example."

Source

What information should it use?

"Use the photo of the textbook page I'm attaching."

Putting it together: a complete prompt

Explain how to solve equivalent fractions so I can help my 10-year-old daughter with her math homework. **Use the photo of her textbook page I'm attaching.** Use simple words and a pizza example so I can explain it to her.

Goal **Context** **Source** **Expectations**

Vague prompt vs. clear prompt

Writing to the teacher

Vague prompt

Write an email to the teacher.

Clear prompt

Help me write a short, respectful email to my son's 3rd-grade teacher. I want to ask for a meeting because my son says he doesn't understand the reading assignments. I can meet Tuesday or Thursday after 4 p.m. Write it in English and also give me a Spanish translation.

More detail = better answers. And you can ask for the answer in two languages.

Tip 1: Give the AI a role

At the start of the conversation, tell the AI who it should be. It adjusts its tone, level, and expertise for the whole session.

No role

"Explain photosynthesis."

With a role

"You are a patient, fun elementary school teacher. Explain photosynthesis to an 8-year-old in 5 sentences."

Tip 2: Ask the AI to guide, not to solve

For homework, AI works better as a tutor than as an answer machine. Ask it to ask questions instead of giving the answer.



"I'm 12 and I don't understand this math problem: [problem]. Don't give me the answer. Ask me questions, one at a time, so I can reach the solution step by step."

This works for any subject: "Don't give me the answer, help me think."

Tip 3: Ask it to check its answer

AI can review its own answer, find a mistake, and fix it. Ask: "Check your answer. Is it correct?"

Question

"Is 17 a prime number? Explain your answer."

AI answer

"17 is not prime because it can be divided by 1, 17, and also by 3."

Self-check

"That answer is wrong. 17 divides evenly only by 1 and 17. So yes, it is prime."

03
Our kids and AI

School, risks, and opportunities

What are our kids doing with AI?



Homework

Ask questions, summarize readings, request whole essays



Chatbot "friends"

Talk with characters or virtual companions



Photos and filters

Edit photos and create images and videos with AI



Games and apps

AI is already built into social media and games

Use AI to learn, not to copy

Academic honesty

Copying

Write me a 5-paragraph essay about the Mexican Revolution for my history class.

Learning

I don't understand what caused the Mexican Revolution. Can you explain it as if I were 10 years old? Then ask me 3 questions to check whether I understood.

Teachers can often tell, but more importantly, the student learns nothing. AI is a great tutor, as long as the student keeps thinking.

Minimum ages and parental controls

ChatGPT

- Minimum age: 13
- Under 18 requires a parent's permission
- Parental controls: link your teen's account, set quiet hours, turn off memory, voice, or images

Other apps

- Gemini, Copilot, and Meta AI (WhatsApp, Instagram) have their own age rules
- Review each app's settings together with your kids
- Keep devices in shared spaces at home

Rules change often: check each app's current settings.

Risks we should know about



Companion chatbots

Characters that act as friends or "virtual partners." They can create emotional dependence and isolate young people.



Fake images and videos

Fake photos or videos (deepfakes) used to deceive, mock, or harass others.



Voice-cloning scams

"Mom, I'm in trouble, send me money." A few seconds of audio can be enough to copy a voice.

Simple protection: agree on a family code word. If someone asks for money urgently, ask for the code word.

How to talk with our kids about AI

Questions to start the conversation at home, without judgment:

> What do you use AI for? Can you show me how?

> Has AI ever given you a wrong answer? How did you notice?

> What do your teachers think about using AI for homework?

> Have you ever talked to a chatbot like it was a friend? How did it feel?

04

Hands-on exercise in pairs

Let's improve a prompt together

Exercise: from vague to clear

Starting point: *"Help me with my son's homework."*

1

Add the goal

Which subject and topic?

2

Add context

How old is the child and what grade?

3

Add expectations

Simple explanation, examples, questions?

4

Try and compare

Write both versions and compare the answers

Time: 10 minutes. Then we'll share 2 or 3 examples.

05

Can we trust AI?

Hallucinations, privacy, and scams

Strengths and weaknesses of generative AI

Strengths

- Creates content quickly
- Adapts to each person
- Translates and summarizes
- Explains patiently, at any hour
- Helps with writing and code
- Improves accessibility

Weaknesses

- Hallucinations: makes things up
- Inconsistent answers
- Bias and unfairness
- Security and privacy risks
- Copyright issues
- Doesn't truly understand what it says

Hallucinations: AI makes things up too

Question

"When did Frida Kahlo die and what was her last painting?"

AI answer

"Frida Kahlo died on July 13, 1954. Her last painting was Viva la Vida, painted just days before her death."

The date is correct, but whether the painting was made "just days before her death" is debated. The AI states it as proven fact.

How would you check this?

Sounding sure is not the same as being right.

Privacy: share the situation, not your data

Don't do this

"My name is María González, my Social Security number is 123-45-6789, and my tourist visa has expired. What should I do?"

Do this instead

"A person's tourist visa has expired. What general options exist?"

Treat the chat like a conversation out loud in public. Free versions may use your conversations to train the model. In ChatGPT: Settings → Data controls → turn off "Improve the model for everyone."

06
AI in everyday life

From a simple prompt to an advanced one

Create a family chore system

Home life

Beginner

We're a family with kids ages 7, 10, and 14. We want to start a chore system. Suggest methods other families use.

Advanced

You are a family coach. We have 3 kids (ages 7, 10, and 14). Simple lists didn't work for us. Suggest 3 or 4 methods for a long-term chore system. For each, explain how to start, which incentives fit each age, and how to keep kids accountable without conflict. No complicated apps.

Prepare for a doctor's visit

Health and wellness

Beginner

I'm going to the doctor soon. How can I prepare for my appointment?

Advanced

You are a medical assistant. I'm 35 and seeing the doctor for frequent headaches that started 3 months ago. I sometimes take ibuprofen, but it doesn't always help. Give me: (1) key questions for the doctor, (2) which symptoms to describe in detail, (3) which tests I could ask about.

AI does not replace your doctor: use it to prepare, not to diagnose yourself.

Practice a job interview

Career development

Beginner

I have a job interview soon. Help me practice.

Advanced

Act as an HR interviewer. I have an interview for a sales associate job at a sportswear store, and I have 2 years of experience. Ask me 5 questions, one at a time. After each answer, tell me what I did well, what I can improve, and how to say it more professionally.

Organize the family budget

Personal finance

Beginner

I want to save more money each month. How can I organize my budget better?

Advanced

Act as a financial advisor. I earn $2,500 a month. Fixed expenses: rent $900, transportation $200, food $400, utilities $150. I have $1,200 in credit card debt. I want to save $200 a month and pay off the debt in 12 months. Give me a monthly plan, which expenses to cut first, and how to pay the debt while still saving.

Use example numbers: never share your real bank details.

Keep the kids entertained

Fun and entertainment

Beginner

I have two kids, ages 6 and 9. We're home this weekend. Give me 5 fun activities that don't cost much.

Advanced

Act as a child development specialist. I have two kids, ages 6 and 9, a small yard, and basic supplies. Give me: (1) a Saturday plan broken down by hour, (2) one creative and one physical activity for each child, (3) one activity to do together. Keep it practical and easy to follow.

Many more ideas for everyday life

Home

Decluttering, home improvement, car maintenance

Food & health

Weekly menus, less food waste, workout plans

Work

Résumés, interviews, professional emails

Money

Budgeting, saving, tax prep, selling things

Learning

Learning English, new skills, studying for exams

Family

Resolving conflicts, planning parties, gift ideas

Travel

Itineraries, packing lists, translation

Creativity

Stories starring your child, poems, games

Free AI tools

ChatGPT

chatgpt.com

Write, ask, explain, and practice languages. Free app with voice mode.

Gemini (Google)

gemini.google.com

Built into Google: search, summaries, and help with Gmail.

Microsoft Copilot

copilot.microsoft.com

Free in the Edge browser, Windows, and the mobile app.

Meta AI

WhatsApp and Instagram

Already inside apps many families use every day.

07

Practical Family Guide

For using ChatGPT and other AI tools at home

Guide: our family rules



We think on our own first

We try the homework before using AI



AI explains; it doesn't do the work

We use it as a tutor and write in our own words



We check what matters

We compare with the textbook, the teacher, or another trusted source



We protect our information

No full names, addresses, photos, or documents



We use it in shared spaces

And we talk as a family about what we find



If it feels wrong, we say so

No punishment: asking is always okay

Guide: what never to share and how to check

Never share

- Full name and address
- Social Security, passport, or visa numbers
- Bank details or passwords
- Photos of your children
- Your child's school name
- Personal medical information

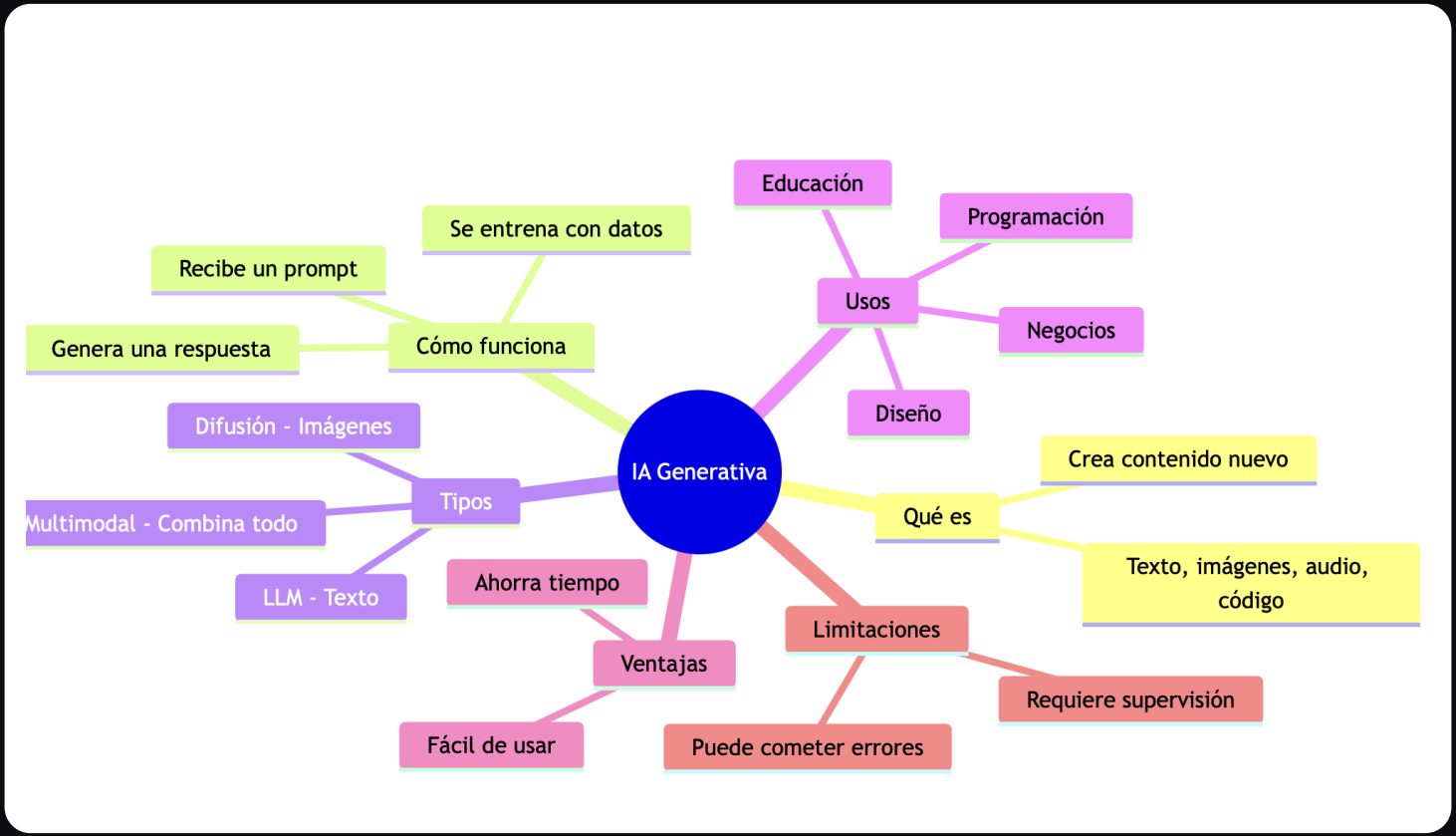
How to check

1. Ask: "What is your source?"
2. Compare with the textbook or the teacher
3. Look up the fact in 2 trusted sources
4. Be wary of dates, numbers, and exact quotes
5. Ask: "Check your answer"

Guide: helpful prompts for homework

1. *"Explain [topic] as if I were [age] years old."*
2. *"Don't give me the answer. Ask me questions so I can get there."*
3. *"Give me a 5-question practice quiz on [topic]."*
4. *"Review my writing and tell me what to improve, without rewriting it."*
5. *"Translate the homework instructions into Spanish so I can understand them."*
6. *"Give me 3 ideas to start my project, not the whole project."*

Conclusions: generative AI at a glance



Mind map in Spanish

Three messages to take home



Check it

AI sounds confident even when it's wrong. Compare with trusted sources.



Protect your data

Describe the situation in general, never your personal details.



Stay involved

Talk, ask questions, and use AI together with your kids to learn.

Back to the scale: where do you feel you are with AI now?

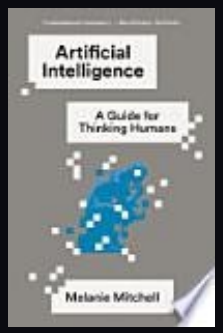
Resources to keep learning

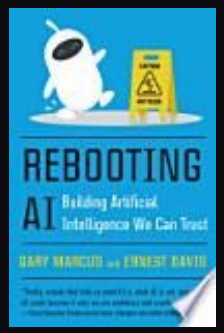
For families

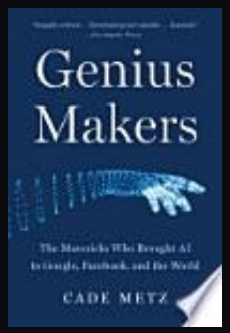
Common Sense Media
commonsensemedia.org
Guides and reviews for families on technology and AI, also in Spanish

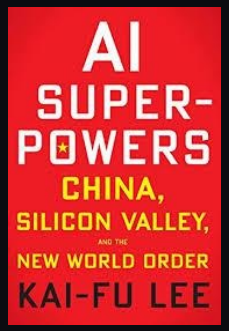
Practical Family Guide
Included in this presentation (section 07)

Books

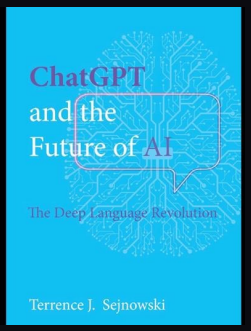












Questions and answers

Thank you for joining us!

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https://github.com/iportilla/BSD

